

Financial Econometrics: Generalized AutoRegressive Conditional Heteroskedasticity (GARCH)

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*Note: This lecture note covers Ch 20 (Nonlinearities in Variance) of
Econometric Modelling with Time Series: Specification, Estimation and Testing
by V. L. Martin, A. S. Hurn and D. Harris.*

Motivation

Conditional Expectation

Conditional vs. Unconditional Risk Measures

Empirical Regularities of Asset Returns Related to Volatility

Engle's ARCH(p) Model

Bollerslev's GARCH Model

Model extensions

Estimating GARCH by MLE

Python Implementation

Outline

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Figure: Robert F. Engle

- ▶ Robert Fry Engle III (born November 10, 1942) is an American economist and statistician.
- ▶ He won the 2003 Nobel Memorial Prize in Economic Sciences, sharing the award with Clive Granger, “for methods of analyzing economic time series with time-varying volatility”. He develops ARCH models.



Figure: Tim Bollerslev

- ▶ Bollerslev, Tim. "Generalized autoregressive conditional heteroskedasticity." *Journal of econometrics* 31.3 (1986): 307-327.
- ▶ In his paper, he introduce GARCH models.

- ▶ **Volatility Clustering:** GARCH models capture periods of high and low volatility clustering observed in financial markets.
- ▶ **Time-Varying Volatility:** They allow for modeling volatility as a dynamic process, reflecting the changing risk and uncertainty in financial assets over time.
- ▶ **Risk Management:** GARCH models are essential for calculating Value at Risk (VaR) and Expected Shortfall, aiding in effective risk assessment and mitigation.

- ▶ **Option Pricing:** By estimating future volatility, GARCH models improve the accuracy of derivative pricing models like Black-Scholes.
- ▶ **Economic Forecasting:** Useful in forecasting economic conditions by modeling the volatility of macroeconomic variables like inflation rates.
- ▶ **Investment Strategies:** They help investors adjust portfolios based on predicted volatilities, informing decisions about asset allocation and hedging.

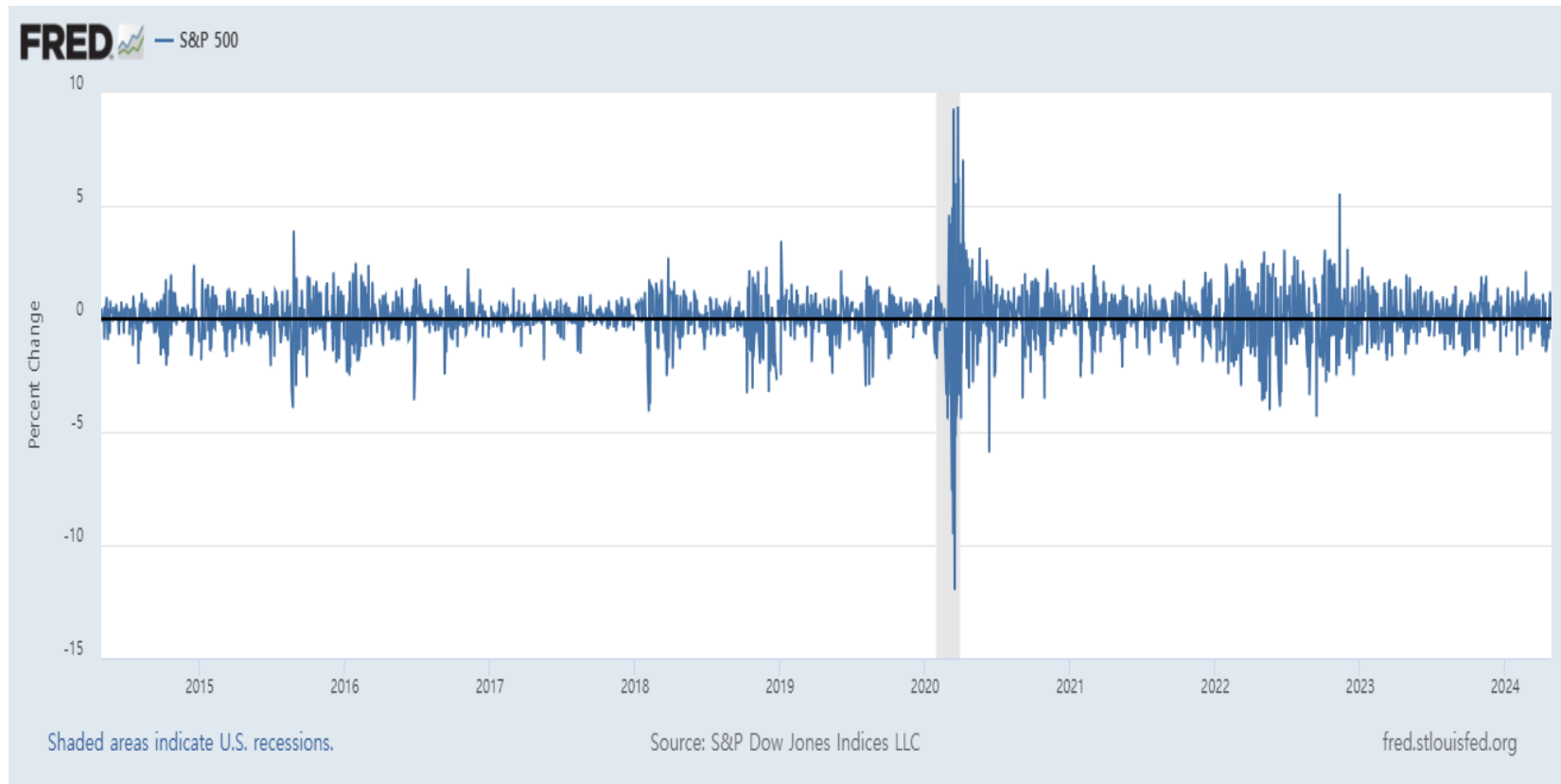


Figure: S&P500 daily return

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Conditional Expectation

Let $\{Y_t\}$ be a sequence of random variables and let $\{I_t\}$ be a sequence of information sets. For example,

$$I_t = \{Y_1, Y_2, \dots, Y_t\} = \text{past history of } Y_t$$

$$\text{or } I_t = \{(Y_s, X_s)_{s=1}^t\}, \{X_t\} = \text{explanatory variables}$$

Conditional Expectation

Conditional Expectation: Let Y_t be a continuous random variable with conditional pdf $f(Y_t | I_s)$, where I_s is an information set upto time s . Then

$$\mu_{t|s} = E[Y_t | I_s] = \int_{-\infty}^{\infty} y_t f(y_t | I_s) dy_t$$

$$\sigma_{t|s}^2 = \text{Var}[Y_t | I_s] = \int_{-\infty}^{\infty} (y_t - E[Y_t | I_s])^2 f(y_t | I_s) dy_t$$

Recap: Conditional Expectation

Property 1: Let X and Y denote random variables and let I denote an information set. If $X \subset I$, then

$$E[XY \mid I] = XE[Y \mid I]$$

Property 2: Law of Iterated Expectation

Let I_1 and I_2 be information sets such that $I_1 \subseteq I_2$, and let Y be a random variable such that $E[Y \mid I_1]$ and $E[Y \mid I_2]$ are defined. Then

$$E[Y \mid I_1] = E[E[Y \mid I_2] \mid I_1] \text{ (smaller set always wins)}$$

Example: If $I_1 = \emptyset$ (empty set) then

$$E[Y \mid I_1] = E[Y] \text{ (unconditional expectation)}$$

$$E[Y] = E[E[Y \mid I_2] \mid I_1] = E[E[Y \mid I_2]]$$

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Unconditional modelling

Let R_{t+1} denote an asset return between times t and $t + 1$.

Unconditional Modelling: Unconditional modelling of R_{t+1} is based on the unconditional or marginal distribution of R_{t+1} . That is, risk measures are computed from the marginal distribution F_R

$$R_{t+1} \sim F_R, E[R_{t+1}] = \mu, \text{var}(R_{t+1}) = \sigma^2$$

Define

$$Z_{t+1} = \frac{R_{t+1} - \mu}{\sigma}, Z_{t+1} \sim F_Z, E[Z_{t+1}] = 0, \text{var}(Z_{t+1}) = 1$$

So that the unconditional model is given by

$$\checkmark R_{t+1} = \mu + \sigma \times Z_{t+1}$$

Conditional modelling

Let I_t denote information known at time t . For example,

$$I_t = \{R_t, R_{t-1}, \dots, R_0\} \text{ or}$$

$$I_t = \{(R_t, X_t), (R_{t-1}, X_{t-1}), \dots, (R_0, X_0)\}.$$

Conditional Modelling: Conditional modelling of R_{t+1} is based on the conditional distribution of R_{t+1} given I_t . That is, risk measures are computed from the conditional distribution $F_{R|I_t}$

$$R_{t+1} \sim F_{R|I_t}, E[R_{t+1} | I_t] = \mu_{t+1|t}, \text{Var}[R_{t+1} | I_t] = \sigma_{t+1|t}^2$$

Define

$$Z_{t+1} = \frac{R_{t+1} - \mu_{t+1|t}}{\sigma_{t+1|t}}, Z_{t+1} \sim F_Z, E[Z_{t+1}] = 0, \text{Var}[Z_{t+1}] = 1$$

So that the conditional model is given by

$$\checkmark R_{t+1} = \mu_{t+1|t} + \sigma_{t+1|t} \times Z_{t+1}$$

Conditional Mean, Variance and Volatility

- ▶ $E[R_{t+1} | I_t] = \mu_{t+1|t}$ = conditional mean
- ▶ $Var[R_{t+1} | I_t] = \sigma_{t+1|t}^2$ = conditional variance
- ▶ $\sigma_{t+1|t}$ = conditional volatility
- ▶ Intuition: As I_t changes over time so does $\mu_{t+1|t}$ and $\sigma_{t+1|t}$
- ▶ Remark: For many daily asset returns, it is often safe to assume $\mu_{t|t-1} = \mu \approx 0$ but it is not safe to assume $\sigma_{t+1|t} = \sigma$
 ~~$\mu_{t|t-1}$~~

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Empirical Regularities

- ▶ **Thick tails:** Excess kurtosis decreases with aggregation (daily return has strong excess kurtosis while monthly return has weak excess kurtosis.)
- ▶ **Volatility clustering:** Large changes followed by large changes; small changes followed by small changes
- ▶ **Leverage effects:** changes in prices often negatively correlated with changes in volatility (after a market crash, a high volatility regime tends to begin.)
- ▶ **Forecastable events:** Forecastable releases of information are associated with high ex ante volatility
- ▶ **Volatility and serial correlation:** Inverse relationship between volatility and serial correlation of stock indices
- ▶ **Volatility co-movements:** Evidence of common factors to explain volatility in multiple series

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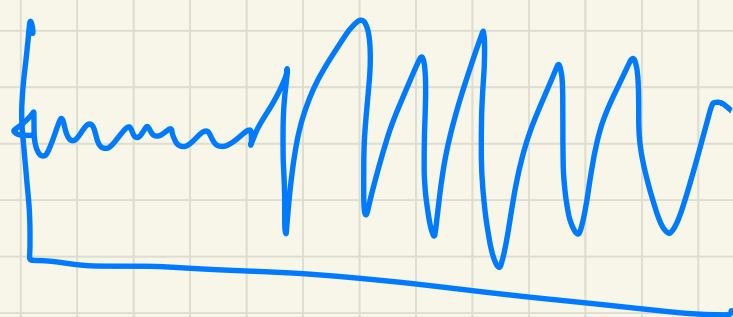
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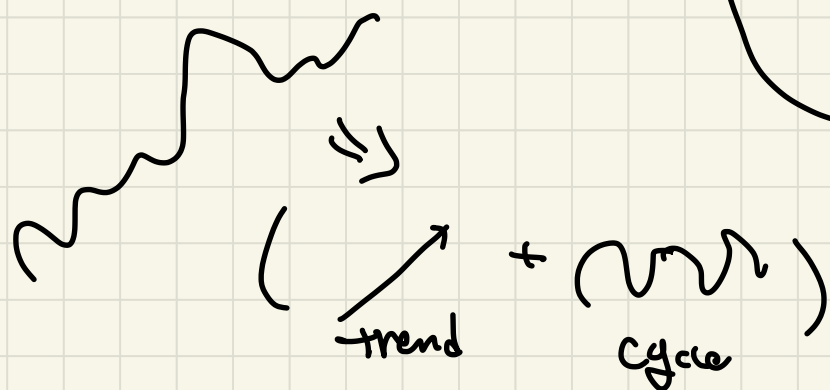
Python Implementation



時間序列

- ① conditional expectation
- ② Law of iterative exp.

$$(1) \quad \underline{Y_t} = X_t' \beta + e_t$$



if, unconditional

$$e_t | I_{t-1} \sim N(0, \sigma_t^2)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 e_{t-1}^2$$

conditional

$$\text{let } E(\sigma_t^2) = \sigma_t^2$$

$$① \quad E(\sigma_t^2) = \alpha_0 + \alpha_1 E(e_{t-1}^2), \quad \sigma_t^2 = \frac{\alpha_0}{1 - \alpha_1}$$

Model specification

Intuition: Use an autoregressive model to capture time dependence in conditional volatility in asset returns

The ARCH(p) model for $r_t = \ln P_t - \ln P_{t-1}$ (log return) is

$$r_t = E[r_t | I_{t-1}] + \epsilon_t, \quad \epsilon_t | I_{t-1} \sim N(0, \sigma_t^2)$$
$$\sigma_t^2 = a_0 + a_1 \underbrace{\epsilon_{t-1}^2} + \cdots + a_p \underbrace{\epsilon_{t-p}^2}, \quad a_0 > 0, a_i \geq 0$$

- Think of why all coefficients are restricted to be positive!

Model specification

Alternative error specification is given by

$$\epsilon_t = \sigma_t z_t,$$

$$z_t \sim i.i.d.N(0, 1),$$

$$\sigma_t^2 = a_0 + a_1 \epsilon_{t-1}^2 + \cdots + a_p \epsilon_{t-p}^2,$$

$$a_0 > 0, a_i \geq 0.$$

- ▶ The random variable z_t doesn't have to be normal. It can have a fat-tailed distribution; e.g. Student's-t distribution
- ▶ If z_t is non-normal, it must be standardized so that $\text{var}(z_t) = E[z_t^2] = 1$.

Properties of ARCH Errors: ARCH(p) Process

= order

$$\epsilon_t = \sigma_t z_t,$$

$$z_t \sim i.i.d.N(0, 1),$$

$$\sigma_t^2 = a_0 + a_1 \epsilon_{t-1}^2 + \cdots + a_p \epsilon_{t-p}^2,$$

= $a_0 > 0, a_i \geq 0.$

- ▶ $\{\epsilon_t\}$ is covariance stationary provided $0 \leq a_1 + \cdots + a_p < 1$.
- ▶ $E[\sigma_t^2] = \bar{\sigma}^2 = \frac{a_0}{1 - a_1 - \cdots - a_p}$

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Model specification

Autoregressive → conditional hetero

Idea: ARCH is like an AR model for volatility. GARCH is like an ARMA model for volatility.

The GARCH(p, q) model is

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim i.i.d.N(0, 1),$$

$$\sigma_t^2 = a_0 + a_1 \epsilon_{t-1}^2 + \dots + a_p \epsilon_{t-p}^2 + b_1 \sigma_{t-1}^2 + \dots + b_q \sigma_{t-q}^2,$$

$$a_0 > 0, a_i \geq 0, b_j \geq 0$$

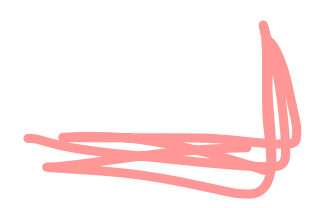
- ▶ For identification of b_j , must have at least one ARCH coefficient $a_i > 0$ for $i > 0$.
- ▶ GARCH (1, 1) is very often the best model

GARCH(1,1)

The most commonly used GARCH(p, q) model is the GARCH(1, 1)

$$\left[\begin{array}{l} \sigma_t^2 = a_0 + a_1 \epsilon_{t-1}^2 + b_1 \sigma_{t-1}^2, \\ a_0 > 0, a_1 \geq 0, b_1 \geq 0. \end{array} \right]$$

- ▶ stationarity condition: $a_1 + b_1 < 1$
- ▶ unconditional variance : $\bar{\sigma}^2 = a_0 / (1 - a_1 - b_1)$



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Market microstructure

Market microstructure refers to the mechanisms and processes that facilitate trading in markets, including how orders are processed and executed, the role of intermediaries, and the impact of the trading system on price formation. Two significant microstructure effects relevant to modeling returns are:

Market microstructure

- ▶ **Bid-Ask Bounce:** This occurs because trades can occur at either the bid or the ask price, causing prices to bounce back and forth, introducing a form of autocorrelation in observed prices (and hence returns). If a stock is bought at the ask price, it may subsequently be sold at the lower bid price, leading to negative autocorrelation in returns.
- ▶ **Non-Trading Effects:** In less liquid stocks, or during periods when trading does not occur frequently, prices may not reflect all available market information immediately. As trades occur, prices adjust to new information, leading to potential autocorrelations as returns "catch up" to where they should be based on new information.

Market microstructure

- ▶ Why ARMA for Microstructure? When $E[r_t | I_{t-1}]$ is modeled as an ARMA process, it can potentially capture the autocorrelations due to microstructure effects:
- ▶ The AR part can model the momentum or continuation in returns due to persistent effects, such as those from the trading process itself or trader behaviors. (e.g., Non-Trading Effects)
- ▶ The MA part can help capture the "noise" introduced by microstructure effects, where the observed return at any time might be influenced by transient factors that do not have a long-term impact. (e.g., bid-ask bounce)

Conditional Mean Specification

- ▶ $E[r_t \mid I_{t-1}]$ is typically specified as a constant or possibly a low order ARMA process to capture autocorrelation caused by market microstructure effects.
- ▶ If extreme or unusual market events have happened during sample period, then dummy variables associated with these events are often added to the conditional mean specification to remove these effects. The general conditional mean specification is of the form

$$E[r_t \mid I_{t-1}] = c + \sum_{j=1}^p \phi_j r_{t-j} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \sum_{j=1}^{\ell} \beta'_j \mathbf{x}_{t-j}$$

where $\mathbf{x}_t$ is a $k \times 1$ vector of exogenous explanatory variables.
(ARMA(p,q) with exogenous variables)

Explanatory Variables in the Conditional Variance Equation

- ▶ Exogenous explanatory variables may also be added to the conditional variance formula

$$\sigma_t^2 = a_0 + \sum_{j=1}^{p'} a_j \epsilon_{t-j}^2 + \sum_{j=1}^{q'} b_j \sigma_{t-j}^2 + \sum_{j=1}^{\ell'} \delta'_j \mathbf{z}_{t-j},$$

where $\mathbf{z}_t$ is a $m \times 1$ vector of variables, and δ is a $m \times 1$ vector of positive coefficients. (GARCH(p' , q') with exogenous variables)

- ▶ Variables that have been shown to help predict volatility are trading volume, interest rates, macroeconomic news announcements, implied volatility from option prices and realized volatility, overnight returns, and after hours realized volatility

Explanatory Variables in the Conditional Variance Equation

- ▶ Overnight returns refer to the price change of a financial asset (such as a stock or commodity) that occurs outside of the regular trading hours.
- ▶ Essentially, overnight returns capture the price movement that happens when the market is closed.
- ▶ Overnight return

$$R_{\text{overnight}} = \frac{P_{\text{open}} - P_{\text{close}}}{P_{\text{close}}}$$

where $R_{\text{overnight}}$ represents the overnight return; P_{open} is the opening price of the asset on the current trading day; P_{close} is the closing price of the asset on the previous trading day.

Explanatory Variables in the Conditional Variance Equation

- ▶ Realized volatility is a measure of the variability of an asset's price, calculated using high-frequency data over a specific period. It provides a direct and empirical estimate of the volatility of an asset by summing the squared returns over that period.

GARCH-in-Mean (GARCH-M)

Idea: Modern finance theory suggests that volatility may be related to risk premia on assets

The GARCH-M model allows time-varying volatility to be related to expected returns

$$r_t = c + \alpha g(\sigma_t) + \epsilon_t$$

$$\epsilon_t \sim \text{GARCH}$$

$$g(\sigma_t) = \begin{cases} \sigma_t \\ \sigma_t^2 \\ \ln(\sigma_t^2) \end{cases}$$

Exponential GARCH (EGARCH)

Idea: Capture asymmetries in volatility, no positive restriction on model parameters

The EGARCH model addresses leverage effects, where negative shocks have a different impact on volatility than positive shocks of the same magnitude.

$$\log \sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \log \sigma_{t-i}^2 + \sum_{j=1}^q g(Z_{t-j}),$$
$$g(Z_t) = \theta Z_t + \gamma(|Z_t| - E[|Z_t|])$$

where:

- ▶ σ_t^2 is the conditional variance (volatility).
- ▶ $Z_t = \frac{\epsilon_t}{\sigma_t}$ and ϵ_t are the standardized residuals from the mean equation.
- ▶ θ and γ capture the asymmetry (according to the sign of ϵ_t) and leverage effect (according to the magnitude of ϵ_t), respectively.

Threshold GARCH (TGARCH)

Idea: Differentiate between positive and negative shocks

The TGARCH model explicitly allows different responses in volatility to positive and negative shocks, often to model leverage effects.

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^p \gamma_i \epsilon_{t-i}^2 I_{\epsilon_{t-i} < 0} + \beta \sigma_{t-1}^2$$

where:

- ▶ ϵ_{t-i} are past shocks.
- ▶ $I_{\epsilon_{t-i} < 0}$ is an indicator function that takes the value 1 if $\epsilon_{t-i} < 0$ (negative shock).
- ▶ γ_i coefficients capture the additional impact of negative shocks compared to positive shocks.

Multivariate GARCH (MGARCH)

Idea: Model volatility across multiple assets

MGARCH models are designed to handle the changing variances and covariances among multiple time series simultaneously.

Common forms of MGARCH include:

- ▶ BEKK
- ▶ Reference of BEKK: Engle, Robert F., and Kenneth F. Kroner. "Multivariate simultaneous generalized ARCH." *Econometric theory* 11.1 (1995): 122-150.
- ▶ DCC (Dynamic Conditional Correlation)
- ▶ Reference of DCC: Engle, Robert. "Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models." *Journal of Business & Economic Statistics* 20.3 (2002): 339-350.

Dynamic Covariance Matrix

- ▶ Let's consider n time series variables (e.g., asset returns), denoted by $r_{1,t}, r_{2,t}, \dots, r_{n,t}$, where t represents time. We can represent these variables at a specific time t as a column vector:

$$\mathbf{r}_t = \begin{pmatrix} r_{1,t} \\ r_{2,t} \\ \vdots \\ r_{n,t} \end{pmatrix}$$

- ▶ The vector of n time series variables can be represented by the sum of the conditional mean vector and error vector as

$$\mathbf{r}_t = E[\mathbf{r}_t | \mathbf{I}_{t-1}] + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t | \mathbf{I}_{t-1} \sim N(0, \boldsymbol{\Sigma}_t)$$

- ▶ The conditional mean vector could be modelled by time series models such as VAR models.

Dynamic Covariance Matrix

- ▶ The dynamic covariance matrix Σ_t is an $n \times n$ matrix whose (i, j) -th element represents the conditional covariance between the i -th and j -th time series variables at time t .
- ▶ The dynamic covariance matrix $\Sigma_t = \text{Var}[\mathbf{r}_t | \mathbf{I}_{t-1}] = \text{Var}[\boldsymbol{\varepsilon}_t | \mathbf{I}_{t-1}]$ can be written as:

$$\Sigma_t = \begin{pmatrix} \sigma_{1,t}^2 & \sigma_{1,2,t} & \cdots & \sigma_{1,n,t} \\ \sigma_{2,1,t} & \sigma_{2,t}^2 & \cdots & \sigma_{2,n,t} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n,1,t} & \sigma_{n,2,t} & \cdots & \sigma_{n,t}^2 \end{pmatrix}$$

where

$$\sigma_{i,t}^2 = \text{Var}[r_{i,t} | I_{t-1}] = E[(r_{i,t} - E[r_{i,t} | I_{t-1}])^2 | I_{t-1}]$$

and

$$\sigma_{i,j,t}^2 = E[(r_{i,t} - E[r_{i,t} | I_{t-1}])(r_{j,t} - E[r_{j,t} | I_{t-1}]) | I_{t-1}]$$

for $i, j = 1, 2, \dots, n$

BEKK Model

Idea: Model covariance among multiple assets with stability guarantees

The BEKK model, a specific form of MGARCH, ensures positive definiteness of the covariance matrix and is formulated as follows:

$$\Sigma_t = \mathbf{C}\mathbf{C}' + \sum_{i=1}^k \mathbf{A}_i \boldsymbol{\varepsilon}_{t-i} \boldsymbol{\varepsilon}_{t-i}' \mathbf{A}_i' + \sum_{j=1}^l \mathbf{B}_j \Sigma_{t-j} \mathbf{B}_j'$$

where:

- ▶ $\mathbf{C}$ is restricted to be a lower-triangular matrix.
- ▶ $\mathbf{H}_t$ is the conditional covariance matrix of returns at time t .
- ▶ $\boldsymbol{\varepsilon}_{t-i}$ are the $(n \times 1)$ residual vector from the mean equation at time $t - i$.
- ▶ $\mathbf{C}, \mathbf{A}_i, \mathbf{B}_j$ are matrices that need to be estimated.
- ▶ This model captures both own-volatilities and cross-covariances dynamically.

Dynamic Conditional Correlation (DCC) Model

Idea: Allow time-varying correlations among multiple assets

The DCC model is highly popular for its flexibility in modeling

time-varying correlations:

$$\Sigma_t = \mathbf{D}_t \mathbf{Q}_t \mathbf{D}_t,$$

where:

$$\mathbf{D}_t = \begin{pmatrix} \sigma_{1,t} & 0 & \cdots & 0 \\ 0 & \sigma_{2,t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{n,t} \end{pmatrix}, \quad \mathbf{Q}_t = \begin{pmatrix} 1 & \rho_{1,2,t} & \cdots & \rho_{1,n,t} \\ \rho_{2,1,t} & 1 & \cdots & \rho_{2,n,t} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n,1,t} & \rho_{n,2,t} & \cdots & 1 \end{pmatrix}$$

- ▶ $\mathbf{D}_t$ is the diagonal matrix of conditional standard deviations.
(estimated by univariate GARCH for each return series)
- ▶ $\mathbf{Q}_t$ is the dynamic correlation matrix.

Dynamic Conditional Correlation (DCC) Model

Idea: Allow time-varying correlations among multiple assets

The dynamic correlation matrix is modeled as:

$$\mathbf{Q}_t = \text{diag}(\mathbf{W}_t)^{-\frac{1}{2}} \mathbf{W}_t \text{diag}(\mathbf{W}_t)^{-\frac{1}{2}},$$
$$\mathbf{W}_t = (1 - \alpha - \beta) \overline{\mathbf{W}} + \alpha(\mathbf{z}_{t-1} \mathbf{z}_{t-1}') + \beta \mathbf{W}_{t-1},$$

where:

- ▶ $\mathbf{z}_{t-1}$ are standardized residuals. ($z_{i,t} = \frac{\varepsilon_{i,t}}{\sigma_{i,t}}$)
- ▶ The DCC model adapts to changing market conditions, making it suitable for risk management and asset allocation.
- ▶ See the details of the ML estimation method for the DCC model in the Section 20.6.3 of Ch 20 (Nonlinearities in Variance) of *Econometric Modelling with Time Series: Specification, Estimation and Testing*.

Constant Conditional Correlation (CCC) Model

Idea: Simplify the correlation structure among assets

The CCC model assumes that correlations between assets are constant over time but allows for individual volatilities to be time-varying:

$$\mathbf{H}_t = \mathbf{D}_t \mathbf{Q} \mathbf{D}_t$$

where:

- ▶ $\mathbf{H}_t$ is the conditional covariance matrix of returns at time t .
- ▶ $\mathbf{D}_t$ is the diagonal matrix of conditional volatilities (standard deviations).
- ▶ $\mathbf{Q}$ is the constant correlation matrix.

Although less flexible than DCC, the CCC model is computationally simpler and useful in settings where correlations are stable.

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Estimating GARCH by MLE

Consider estimating the following model

$$r_t = E[r_t | \mathbf{I}_{t-1}] + \epsilon_t = \mathbf{x}'_{t-1}\boldsymbol{\beta} + \epsilon_t,$$

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim i.i.d.N(0, 1),$$

$$\sigma_t^2 = a_0 + a_1\epsilon_{t-1}^2 + \dots + a_p\epsilon_{t-p}^2 + b_1\sigma_{t-1}^2 + \dots + b_q\sigma_{t-q}^2$$

- ▶ The regression parameters $\boldsymbol{\beta}$ and GARCH parameters $\boldsymbol{\gamma} = (a_0, a_1, \dots, a_p, b_1, \dots, b_q)'$ can be estimated separately if z_t follows a normal distribution. Theoretically, we can show that it does not matter to jointly or separately estimate the model parameters.
- ▶ Step 1: Estimate $\boldsymbol{\beta}$ by OLS ignoring ARCH errors and form residuals $\hat{\epsilon}_t = r_t - \mathbf{x}'_{t-1}\hat{\boldsymbol{\beta}}$
- ▶ Step 2: Estimate ARCH process for residuals $\hat{\epsilon}_t$ by MLE.

GARCH Likelihood Function Under Normality

Assume $E[r_t | I_{t-1}] = 0$. Let $\boldsymbol{\theta} = (a_0, a_1, \dots, a_p, b_1, \dots, b_q)'$ denote the parameters to be estimated. Since $\epsilon_t = \sigma_t z_t$

$$f(\epsilon_t | I_{t-1}; \boldsymbol{\theta}) = (2\pi\sigma_t^2)^{-1/2} \exp \left\{ -\frac{1}{2\sigma_t^2} \epsilon_t^2 \right\}$$

For a sample of size T , the prediction error decomposition gives

$$\begin{aligned} & f(\epsilon_T, \epsilon_{T-1}, \dots, \epsilon_1; \boldsymbol{\theta}) \\ &= \left(\prod_{t=m+1}^T f(\epsilon_t | I_{t-1}; \boldsymbol{\theta}) \right) \cdot f(\epsilon_1, \dots, \epsilon_m; \boldsymbol{\theta}) \\ &= \left(\prod_{t=m+1}^T (2\pi\sigma_t^2)^{-1/2} \exp \left\{ -\frac{1}{2\sigma_t^2} \epsilon_t^2 \right\} \right) \cdot f(\epsilon_1, \dots, \epsilon_m; \boldsymbol{\theta}) \end{aligned}$$

GARCH Likelihood Function Under Normality

- ▶ $\sigma_t^2 = a_0 + a_1\epsilon_{t-1}^2 + \dots + a_p\epsilon_{t-p}^2 + b_1\sigma_{t-1}^2 + \dots + b_q\sigma_{t-q}^2$ is evaluated recursively given θ and starting values for σ_t^2 and ϵ_t^2 .
- ▶ For example, consider GARCH(1, 1)

$$\sigma_1^2 = a_0 + a_1\epsilon_0^2 + b_1\sigma_0^2$$

given the initial values, ϵ_0^2, σ_0^2 .

- ▶ Need to specify starting values values ϵ_0^2 and σ_0^2 . Then all other σ_t^2 values can be calculated recursively.
- ▶ The log-likelihood function is

$$\begin{aligned} & -\frac{(T-m)}{2} \ln(2\pi) - \sum_{t=m+1}^T \left[\frac{1}{2} \ln(\sigma_t^2) + \frac{1}{2} \frac{\epsilon_t^2}{\sigma_t^2} \right] \\ & + \ln(f(\epsilon_1, \dots, \epsilon_m; \theta)) \end{aligned}$$

GARCH Likelihood Function Under Normality

- Problem: the marginal density for the initial values $f(\varepsilon_1, \dots, \varepsilon_m; \theta)$ does not have a closed form expression so exact MLE is not possible. In practice, the initial values $\varepsilon_1, \dots, \varepsilon_m$ are set equal to zero and the marginal density $f(\varepsilon_1, \dots, \varepsilon_m; \theta)$ is ignored. This is conditional MLE.

GARCH Likelihood Function Under Normality

- ▶ Once the log-likelihood is initialized, it can be maximized using numerical optimization techniques such as a Newton-Raphson optimization algorithm.
- ▶ The numerical optimization algorithm is built in Python or any other popular statistical computer packages.

Outline

Motivation

Conditional Expectation

Conditional vs. Unconditional Risk Measures

Empirical Regularities of Asset Returns Related to Volatility

Engle's ARCH(p) Model

Bollerslev's GARCH Model

Model extensions

Estimating GARCH by MLE

Python Implementation

GARCH estimation in Python

Before running the code, you need to install 'arch' package.

```
1 # import necessary packages and libraries
2 import numpy as np
3 from arch import arch_model
4
5 # Specify the GARCH model: GARCH(1,1)
6 # note: y is the demeaned return data
7 model = arch_model(y, p=1, q=1)
8
9 # Fit the model
10 results = model.fit()
11
12 # Display the model fitting summary
13 print(results.summary())
```

The End. More python codes
for GARCH extensions are
provided in the accompanied
Jupyter notebook!